Congruences and last digits

High school · number theory · Synthesis problem · 03

> **Problem · Investigate, justify, conclude.** An integer N is congruent to 2 modulo 3 and to 3 modulo 5. Write $N = 3k + 2$, determine k modulo 5, then give all solutions modulo 15. Which solutions from 0 to 100 inclusive are even?